KevaBook Requirements Specification

1. Introduction

1.1 Project Overview

KevaBook is a cloud-based booking platform designed to simplify appointment scheduling and booking management for freelancers, tutors, consultants, coaches, and small service-based businesses.

The platform enables business owners to create professional business profiles, define the services they offer, manage their availability, and receive bookings from clients through a centralized online platform. Clients can discover businesses, view available services, and book appointments without the need for manual scheduling through phone calls, emails, or messaging applications.

KevaBook follows a multi-tenant model, allowing multiple independent businesses to operate within the same platform while maintaining their own business information, services, schedules, and bookings.

The system is intended to reduce administrative overhead for service providers, improve booking efficiency, and provide clients with a convenient self-service booking experience.

1.2 Problem Statement

Many freelancers and small service-based businesses rely on manual appointment scheduling through phone calls, emails, social media messages, or messaging applications such as WhatsApp. These approaches often lead to scheduling conflicts, missed appointments, double bookings, and inefficient communication between businesses and clients.

Additionally, many small businesses lack the resources to develop and maintain their own booking systems.

KevaBook addresses these challenges by providing a centralized platform where businesses can manage their services and appointments while allowing clients to book available time slots online.

1.3 Project Goals

The primary goals of KevaBook are:

- Simplify appointment scheduling for service-based businesses.
- Enable businesses to manage services and availability online.
- Allow clients to book appointments through a self-service platform.
- Reduce scheduling conflicts and double bookings.
- Automate booking notifications and reminders.
- Provide a scalable foundation for future growth and feature expansion.

1.4 Target Users

The platform targets the following users:

Business Owners

Business owners are individuals or organizations that provide services and wish to accept bookings online.

Examples include:

- Tutors
- Consultants
- Freelancers
- Coaches
- Therapists
- Accountants
- Photographers
- Trainers

Clients

Clients are individuals seeking services from businesses registered on the platform.

Clients can:

- Browse available services
- View business profiles
- Book appointments
- Receive booking confirmations and notifications

1.5 Scope of MVP

The Minimum Viable Product (MVP) will focus on the core booking workflow.

Included Features:

- User registration and authentication
- Business profile management
- Service management
- Availability management
- Appointment booking
- Booking management
- Email notifications
- Sms notifications

Excluded from MVP:

- Online payments
- Ratings and reviews
- Video conferencing integration
- Calendar synchronization
- Analytics dashboards
- Team/staff management
- Subscription billing

2. Functional Requirements

2.1 User Management

FR-01 User Registration

The system shall allow users to create an account using an email address and password.

FR-02 User Authentication

The system shall allow registered users to log in using valid credentials.

FR-03 Password Reset

The system shall allow users to reset their password through email verification.

FR-04 User Profile Management

The system shall allow users to view and update their profile information.

2.2 Business Management

FR-05 Business Profile Creation

The system shall allow authenticated users to create a business profile.

FR-06 Business Profile Update

The system shall allow business owners to update business information.

Business information may include:

- Business name
- Description
- Contact details
- Location

- Business logo

FR-07 Business Profile Viewing

The system shall allow clients to view business profiles.

2.3 Service Management

FR-08 Service Creation

The system shall allow business owners to create services.

Each service shall contain:

- Service name
- Description
- Duration
- Price
- Category

FR-09 Service Update

The system shall allow business owners to update service details.

FR-10 Service Removal

The system shall allow business owners to deactivate or remove services.

FR-11 Service Listing

The system shall allow clients to view available services offered by a business.

2.4 Availability Management

FR-12 Availability Configuration

The system shall allow business owners to define working days and working hours.

FR-13 Availability Update

The system shall allow business owners to modify their availability schedule.

FR-14 Slot Generation

The system shall automatically generate available booking slots based on defined availability and service duration.

2.5 Booking Management

FR-15 Service Booking

The system shall allow clients to book an available service.

FR-16 Booking Confirmation

The system shall generate a booking record after successful booking.

FR-17 Booking Viewing

The system shall allow business owners to view bookings associated with their business.

FR-18 Booking Cancellation

The system shall allow bookings to be cancelled.

FR-19 Double Booking Prevention

The system shall prevent multiple bookings from being created for the same time slot.

FR-20 Booking Status Management

The system shall maintain booking statuses including:

- Pending

- Confirmed
- Cancelled
- Completed

2.6 Search and Discovery

FR-21 Business Search

The system shall allow clients to search for businesses.

FR-22 Service Search

The system shall allow clients to search for services.

FR-23 Business Service Viewing

The system shall allow clients to browse services offered by a selected business.

2.7 Notification Management

FR-24 Booking Confirmation Notification

The system shall send an email and sms notification when a booking is successfully created.

FR-25 Booking Cancellation Notification

The system shall send an email and sms notification when a booking is cancelled.

FR-26 Reminder Notification

The system shall send appointment reminder notifications before scheduled appointments.

3. Non-Functional Requirements

3.1 Performance

NFR-01 Response Time

The system shall process and respond to 95% of API requests within 500 milliseconds under normal operating conditions.

NFR-02 Booking Processing Time

The system shall create and confirm bookings within 1 second under normal operating conditions

NFR-03 Concurrent Users

The system shall support at least 100 concurrent users while maintaining the response time requirements defined in NFR-01 and NFR-02.

3.2 Scalability

NFR-04 Horizontal Scalability

The system shall support horizontal scaling of services as demand increases.

NFR-05 Service Independence

System components shall be deployable independently to support future expansion.

3.3 Availability

NFR-06 System Availability

The platform should maintain high availability and minimize service interruptions.

NFR-07 Fault Isolation

Failures in one service should not cause the entire platform to become unavailable.

3.4 Security

NFR-08 Authentication

Access to protected resources shall require user authentication.

NFR-09 Authorization

Users shall only access resources they are permitted to manage.

NFR-10 Data Protection

Sensitive information shall be protected during transmission and storage.

NFR-11 Password Security

User passwords shall be stored using secure hashing algorithms.

3.5 Reliability

NFR-12 Data Consistency

The system shall maintain consistent booking records.

NFR-13 Booking Integrity

The system shall ensure that booking transactions are processed accurately and without duplication.

NFR-14 Availability Accuracy

The system shall prevent double booking of appointments slots and maintain booking consistency under concurrent booking requests.

3.6 Maintainability

NFR-15 Modular Design

The system shall follow a modular microservices architecture to facilitate maintenance and future development.

NFR-16 Code Quality

The system shall follow established coding standards and development best practices.

3.7 Usability

NFR-17 User-Friendly Interface

The platform shall provide an intuitive and easy-to-use interface for both business owners and clients.

NFR-18 Mobile Accessibility

The platform shall be accessible through modern desktop and mobile web browsers.

3.8 Observability

NFR-19 Logging

The system shall maintain application logs for monitoring and troubleshooting.

NFR-20 Monitoring

The system shall provide health monitoring for deployed services.

NFR-21 Distributed Tracing

The platform should support distributed request tracing across microservices.